

Aryan Sachdeva

Full-Stack Developer — Machine Learning Engineer
B.Tech in Chemical Engineering, IIT Jodhpur

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in LinkedIn GitHub Portfolio

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, JavaScript

Backend & Frameworks: Django, Django REST Framework, FastAPI, React, Tailwind CSS

AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, RAG, Vector Search, Sentence Transformers, LLMs

Cloud & Tools: PostgreSQL, FAISS, Redis, Git, GitHub, AWS, Render, Vercel

WORK EXPERIENCE

PHC Booking System

Jan 2025 – Present

React, Django, Django REST Framework, PostgreSQL, JWT

Design Credit — Under Prof. Romi Banerjee

- Architected a decoupled full-stack healthcare booking application with a React SPA frontend and a robust Django REST Framework backend API.
- Mitigated booking race conditions during high-traffic periods by implementing atomic database transactions and row-level locking using `select_for_update()`, completely preventing double-booking of medical slots.
- Optimized database performance by engineering a normalized relational schema and utilizing DRF's `select_related` and `prefetch_related` to eliminate N+1 query inefficiencies.
- Secured API endpoints with JWT authentication, custom RBAC permission classes, and frontend token refresh handling using Axios interceptors and React Hooks.

TECHNICAL PROJECTS

AI-Powered RAG Document Assistant

GitHub

FastAPI, React, FAISS, Sentence Transformers, PostgreSQL

- Built a Retrieval-Augmented Generation (RAG) system for semantic question answering over uploaded PDF documents with citation-grounded responses.
- Implemented hybrid retrieval using BM25 keyword search and FAISS vector similarity with Sentence Transformers embeddings.
- Developed FastAPI REST APIs and a React frontend for semantic search and citation-based PDF navigation.

Movie Recommendation Web App

Live Demo

Django, Django REST Framework, React, Scikit-learn

- Built an end-to-end ML-powered movie recommendation platform with REST API integration.
- Performed data preprocessing, feature extraction, and cosine similarity computation using NumPy, Pandas, and Scikit-learn.
- Developed Django REST APIs and a React frontend, and deployed the application using Render and Vercel.

EDUCATION

Indian Institute of Technology, Jodhpur

2024 – 2028

B.Tech in Chemical Engineering

CGPA: 8.03 / 10

Montfort Sr. Sec. School, Roorkee

2022 – 2024

Senior Secondary (Class XII)

Aggregate: 95.6%

Montfort Sr. Sec. School, Roorkee

2020 – 2022

Secondary (Class X)

Aggregate: 93.33%

RELEVANT COURSEWORK

- Data Structures & Algorithms — Grade: A
- Machine Learning — Grade: A
- Probability and Statistics — Grade: A-